25900138 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCMAT100 - A GATEWAY TO MATHEMATICS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer type Questions

Answer any **5** questions

Each question carries **2** marks

1. If $f(x,y) = 10x^3 + 12y^4$ then find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$. [U] / [CO1]
2. Define critical points of a function of two variables with an example. [U] / [CO1]
3. Define regular partition of an interval. [A] / [CO2]
4. Find $\frac{d}{dx}[\int_a^x f(t)dt]$. [U] / [CO2]
5. How many rows and columns does the coefficient matrix have, if there are m equations in n unknowns? [A] / [CO3]
6. Define an inconsistent system of equations. [U] / [CO3]
7. Define the adjacency matrix of a graph. [K] / [CO4]
8. How many edges does W_8 have? [A] / [CO4]

[2x5 = 10]

Part B

Short Essay Type Questions

Answer any **5** questions

Each question carries **6** marks

9. Find $\frac{dy}{dx}$ at $(0, \ln 2)$ if $xe^y + \sin xy + y - \ln 2 = 0$. [U] / [CO1]

10. Using the chain rule, find $\frac{\partial T}{\partial r}$ and $\frac{\partial T}{\partial \theta}$ if $T = x^2y - xy^3 + 2,\ x = r\cos\theta$ and $y = r\sin\theta$. [U] / [CO1]

11. Evaluate $\int_{-1}^{1}\int_{0}^{2}(4 - x^2 - y^2)\,dy\,dx$. [A] / [CO2]

12. Evaluate $\int_{0}^{\pi/3}\int_{0}^{\pi/6}\sin(2x)\cos(3y)\,dy\,dx$. [A] / [CO2]

13. Using the Gauss elimination method to find the nature of the solution: [A] / [CO3]

$$\begin{aligned} x - y + z &= 4 \\ x - 2y - 2z &= 9 \\ 2x + y + 3z &= 1 \end{aligned}$$

14. Find the solution of the linear system of equations using the Gauss elimination method [A] / [CO3]

$$\begin{aligned} x + y + z &= 3 \\ x + 2y + 3z &= 4 \\ x + 4y + 9z &= 6 \end{aligned}$$

15. How many vertices and edges $K_{r,s}$ will have? [U] / [CO4]

16. Show that a simple graph with at least two vertices must contain two or more vertices of the same degree. [An] / [CO4]

[6x5 = 30]

Part C

Essay Type Questions

Answer any **3** questions

Each question carries **10** marks

17. Find the absolute maximum and minimum value of $f(x, y) = x^2 + y^2$ on the triangular region in the first quadrant bounded by the line $x = 0,\ y = 0$ and $y + 2x = 2$. [U] / [CO1]

18. (a) State the Fundamental Theorem of Calculus, Part 1. [A] / [CO2]

(b) Find $\int_0^4 f(x)\,dx$ if $f(x) = \begin{cases} \sqrt{x}, & 0 \le x < 1 \\ \frac{1}{x^2}, & 1 \le x \le 4 \end{cases}$.

19. Use properties of definite integrals and appropriate formulas from geometry to evaluate the integral $\int_0^1 (5 - 3\sqrt{1 - x^2})dx$. [A] / [CO2]

20. Solve the following system of equations using Gauss elimination and back substitution. [A] / [CO3]

(a) $$\begin{aligned} 2x + 5y &= 2 \\ 4x - 3y &= 30 \end{aligned}$$

(b) $$\begin{aligned} x - y + z &= 0 \\ -x + y - z &= 0 \\ 10y + 25z &= 90 \\ 20x + 10y &= 80 \end{aligned}.$$

21. (a) Define an edge-deleted subgraph and a vertex-deleted subgraph of a graph. Illustrate your definition using examples. [U] / [CO4]

(b) Draw the graphs $C_6 - v$ and $C_6 - e$ where $v \in V(C_6) \quad \& \quad e \in E(C_6)$.

(c) Is C_6 a subgraph of C_6?

22. (a) Draw a non-simple graph with vertex set [U] / [CO4]
$V = \{A, B, C, D, E, F, G, H\}$ and edge set
$E = \{AA, CC, FF, AB, BC, CD, DE, EF, FG, GH, HA\}$.

(b) Identify and list all loops in the graph.

[10x3 = 30]